Laurent **Colbois**

Computer Vision Researcher · PhD in Machine Learning & Biometrics

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“To create a positive impact through technology while following sustainable practices.”

Summary

Computer Vision researcher with PhD from UNIL/Idiap and Master’s from EPFL, specializing in deep learning, Vision-Language Models, and large-scale model training on HPC infrastructure. Recent work benchmarking state-of-the-art VLMs (Gemma3, Qwen2.5-VL, InternVL3) for explainable face recognition. Strong foundation in neural network architectures (CNNs, GANs, Transformers), reproducible ML workflows, and cross-functional collaboration. Proven track record in scientific research with multiple peer-reviewed publications. Eager to apply computer vision expertise to video AI challenges and expand skills in edge deployment and embedded optimization.

Work Experience

Idiap Research Institute — *Martigny, Switzerland*

Postdoctoral Researcher - Vision-Language Models for Biometrics — *Apr. 2025 - Mar. 2026*

- Benchmarked state-of-the-art open-weight Vision-Language Models (Gemma3, Qwen2.5-VL, InternVL3) for explainable face recognition, designing evaluation methodologies for multimodal AI systems
- Trained and scaled large models on institutional HPC infrastructure using Slurm clusters with distributed GPU computing, implementing efficient inference pipelines with vLLM
- Built reproducible ML workflows using modern deep learning stack (PyTorch, Huggingface Transformers) and MLOps tools (Hydra for configuration, Weights & Biases for experiment tracking, MLFlow for model management)
- Led cross-functional coordination of forensic data collection platform, managing stakeholder communication and translating technical requirements across research, development, and legal teams
- Won first place (15K CHF) at 9-day startup hackathon developing AI-based music generation system, demonstrating rapid prototyping and presentation skills to diverse audiences

Idiap Research Institute — *Martigny, Switzerland*

PhD Student - Deepfakes, Biometrics, Computer Vision — *Jan. 2021 - Mar. 2025*

- Developed deep learning models for face morphing attack detection and attribution using CNNs and GANs, achieving state-of-the-art results published in peer-reviewed conferences (IJCB, BIOSIG, ICASSP)
- Designed and implemented neural network architectures for computer vision tasks, including adversarial attack synthesis, detection systems, and generative models
- Created and published synthetic face recognition dataset using GANs, contributing to community benchmarking resources and demonstrating expertise in data generation and augmentation
- Supervised four master’s students on machine learning projects, providing technical mentorship and fostering collaborative research environment
- Taught annual course on forensic evaluation methods, translating complex ML concepts for diverse audiences including forensic scientists and practitioners
- Established reproducible research workflows emphasizing version control, experiment tracking, and code quality—serving as technical resource for lab members

Idiap Research Institute — *Martigny, Switzerland*

Research Intern - Synthetic Data for Face Recognition — *Jul. - Dec. 2020*

- Developed synthetic face dataset generation pipeline using GANs for benchmarking face recognition systems

Logitech — *Lausanne, Switzerland*

Research Intern - Audio Machine Learning — *Feb. - Aug. 2019*

- Designed and implemented automated audio scene classification model for intelligent auto-equalization using deep learning
- Deployed real-time inference demo on embedded microcontroller, gaining initial exposure to on-device ML deployment

Education

University of Lausanne / Idiap Research Institute *Lausanne, Switzerland*

PhD in Forensic Science (Machine Learning & Biometrics) *2021 - 2025*

- Thesis: Face morphing attacks in the era of deepfakes: risks, detection & source attribution

EPFL *Lausanne, Switzerland*

M.Sc. in Computational Science & Engineering *2017 - 2020*

- Thesis: Trainable filterbanks for end-to-end speech enhancement via time-frequency masking
- Skills: Numerical simulations, Machine learning, Deep learning, Signal processing, High performance computing

EPFL *Lausanne, Switzerland*

B.Sc. in Physics *2013 - 2016*

- Skills: Applied mathematics, Numerical physics, Programming, Scientific computing

1 Publications

2026	**Conference Paper (Under Review)**, Benchmarking the Explanatory Quality of Open-Weight Vision-Language Models in Face Recognition	*Face and Gesture*
2024	**Conference Paper**, Evaluating the effectiveness of attack-agnostic features for morphing attack detection	*IJCB*
2023	**Conference Paper**, Approximating Optimal Morphing Attacks using Template Inversion	*IJCB*
2022	**Conference Paper**, On the detection of morphing attacks generated by GANs	*BIOSIG*
2022	**Conference Paper**, Are GAN-based morphs threatening face recognition?	*ICASSP*
2021	**Conference Paper**, On the use of automatically generated synthetic image datasets for benchmarking face recognition	*IJCB*

Skills

Domains	Computer Vision, Vision-Language Models, Deep Learning, Machine Learning, Biometrics, Generative AI (GANs), Explainable AI, Scientific Research
Neural Networks	CNNs, GANs, Transformers, Vision-Language Models, Supervised/Unsupervised Learning, Model Training & Evaluation
ML & Programming	Python (Advanced), PyTorch, Huggingface Transformers, vLLM, TensorFlow (Basic), Scikit-Learn, Pandas, NumPy, C++ (Basic), MATLAB
Infrastructure & MLOps	HPC/Slurm Clusters, Distributed GPU Computing, Docker, Apptainer/Singularity, Git, Conda, Pip, Pixi, Weights & Biases, MLFlow, Hydra, Snakemake, Prefect, Unix/Linux
Research & Collaboration	Scientific Publications (6 peer-reviewed papers), Technical Writing, Cross-functional Communication, Mentorship, Data Collection & Augmentation, Reproducible Workflows
Languages	French (Mother tongue), English (Professional proficiency), German (B2)